Grade: Check Plus

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STERN SCHOOL OF BUSINESS
NEW YORK UNIVERSITY

EXECUTIVE MBA
DC CLASS OF 2027

COR1.1205.85
Statistics and Data Analysis

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ASSIGNMENT 1

Attached Problem.

Due: February 24, 2026 before 6 PM

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The probability distribution of the amount drawn from an ATM at a convenience store is tabulated below. In this table X = the amount withdrawn by a randomly selected customer and f(x) = P[X = x]. A customer can only withdraw the amounts indicated in the first column of the table below.

x	f(x)
10	0.10
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

a. What is the probability that a randomly selected customer withdraws at least $60 from the ATM?

b. Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27.

c. The manager of this convenience score tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager?

d. What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean?

e. What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?

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Assigment/ Problem 1

A	B	C	D
x	f(x)	x · f(x)	$(x-8)^2 f(x)$
10	0.1	1	160
20	0.15	3	135
40	0.30	12	30
60	0.15	9	15
80	0.25	20	225
100	0.05	5	125
		50	690

✓ a.) ✓ Qa probability of withdrawing at least \$60 from ATM

$$P(A \geq 60) = f(60) + f(80) + f(100)$$
$$= 0.15 + 0.25 + 0.05 = 0.45$$

✓ Qb

✓ b.) the mean/weighted average is calculated by x · f(x) as seen in column c above. this is 50

the variance is calculated by $(x-8)^2 f(x)$ as seen in column d above. This is 690. the square root of this is then taken to get standard deviation which is \$26.27

✓ Qc

c.) this is a weighted average so describes what, on average, you can expect withdrawn by any given customer and will describe the general trend over multiple customers.

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Problem/Assignment #1

✓ Qd

d.) $P(50-26.27 \leq A \leq 50+26.27) =$
$P(23.73 \leq A \leq 76.27)$

to whole #s $\Rightarrow$ $24 \leq A \leq 75$ ★can only withdraw specific amounts★

$\Downarrow$ ★

✓ Qe

$40 \leq A \leq 60 = f(40) + f(60) = 0.3 + 0.15 = .45$

e.) $P(50-52.54 \leq A \leq 50+52.54$
$P(-2.54 \leq A \leq 102.54)$ $\Rightarrow$ all numbers fit in so 1.0 or 100%